DDTA - Scheda di lettura compilata

SRC-0042 - BA0-R systems-modeling prior art

REVIEW-ONLY: da approvare prima del caricamento nel repository.

Identita bibliografica e accesso

Source ID	SRC-0042
Titolo	NASA Systems Engineering Handbook, Revision 2
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Anno	2017
Identificatore	NASA/SP-2016-6105 Rev 2; NTRS 20170001761
Accesso	internet_public_official_nasa_pdf
Verification state	verified_full_text_public

1. Research problem and contribution

Provides NASA lifecycle systems-engineering guidance across design, realization and technical management. It treats architecture as an interrelated structure of parts, relationships and connections and identifies multiple representational entities rather than one universal graph edge.

2. Starting artifacts and generated representations

Logical decomposition links functions and relationships to requirements before candidate solution selection. Architecture may be represented by functions, functional flows, interfaces, relationships, resource-flow items, physical elements, containers, modes, links and communication resources. Operational scenarios include events, actions, stimuli, information and interactions.

3. Traceability, change, automation and human role

Interface management explicitly identifies internal/external interfaces, origin, destination, stimuli and characteristics, then captures interface decisions, rationale, assumptions, anomalies and approved changes. Interfaces evolve as architecture emerges and are configuration-managed. Technical data management controls work products, decisions, assumptions, versions and access.

4. Findings, limitations and relationship with DDTA

The “Boundary of Technical Effort” is strong non-security prior art for a responsibility/control boundary: inside is controlled by the technical team, outside influences/is influenced but is not controlled. This supports distinguishing responsibility/system boundary from trust boundary. Architecture terminology is intentionally broad and practice-oriented, so it does not by itself establish a minimal ontology.

Evidenza utile per BA0/BA1

Pag.	Ruolo	Sezione	Interpretazione DDTA
62	foundation	4.3 Logical Decomposition	BA can contain implementation-independent system semantics grounded in requirements.
104	foundation	5.3 Product Verification Guidance	Behavioral representation needs more than static structure.
105	foundation	Figure 5.3-2 End-to-End Data Flow	Flow is a distinct analyzable relation across heterogeneous structural elements.
145	foundation	6.3 Interface Management	Responsibility and interface boundaries are generic engineering concerns.

Pag.	Ruolo	Sezione	Interpretazione DDTA
146	foundation	6.3.1.2 Interface Management Activities	Interface identity/change deserves explicit consideration rather than being inferred only from interaction text.
147	foundation	6.3.1.3 Outputs	Architecture/interface provenance and change history are established engineering needs.
186	challenge	Appendix B Architecture	Actor+Asset+Interaction is far too small as a generic system-modeling vocabulary.
236	foundation	Appendix J 3.5 Boundary of Technical Effort	Strong evidence for responsibility/control boundary distinct from security trust.

Bibliography mining / follow-up

- NASA-HDBK-1009A
- ISO/IEC/IEEE 15288
- INCOSE Systems Engineering Handbook

Governance note

Questa scheda e un ausilio di review. Le future citazioni di tesi devono risolversi alla fonte registrata e a una posizione esatta nell'excerpt ledger approvato. Nessun concetto esterno diventa automaticamente un BAE DDTA.